Research Summary

Graph-Theoretic Analysis of Earthquake Cluster Causal Structure

Bachelor's Thesis — Work in Progress

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1. Research Background and Motivation

Earthquake catalogs record time, location, depth, and magnitude for every detected event. Standard analysis methods — Gutenberg-Richter statistics, Omori-Utsu aftershock decay, and spatial density maps — treat events as individual observations and aggregate over populations. They discard the **event-to-event causal relationships** that encode how one earthquake triggers the next.

Stochastic declustering methods — in particular the ETAS-based approach of Zhuang et al. (2002) and the Model-Independent Stochastic Declustering (MISD) of Marsan & Lengliné (2008) — estimate a matrix of **triggering probabilities w_{ij}**: the probability that event j was triggered by event i. This matrix implicitly defines a directed graph over the catalog, but its topological structure has not been systematically analysed. My thesis develops this graph-theoretic framework and applies it to the Japan earthquake catalog.

2. Dataset

I assembled a merged Japan earthquake catalog from three sources, implementing a custom deduplication pipeline in Python:

- USGS Japan catalog 2001–2018 (CSV, 22-column USGS format)
- Multi-file USGS query exports covering 2000–2025 (ZIP archive, same schema)
- Homogeneous Mw catalog — Table S1 (XLSX, different schema; cross-deduplicated on rounded time/lat/lon/depth key)

Property	Value
Total events	81,369
Time span	1905 – 2025
Dominant source	Homogeneous Mw catalog (63,298 events, 77.8%)
Events M ≥ 4.5	18,553
Events M ≥ 6.0	465
Events M ≥ 7.0	47
Maximum magnitude	M9.1 (2011 Tōhoku)
Columns	25 (time, lat, lon, depth, mag, magType, Mw, Mw_Std, source_file, …)

The deduplication strategy uses authoritative USGS event IDs for USGS-sourced records, and a fuzzy key (time rounded to 1 minute, latitude/longitude rounded to 0.01°, depth rounded to 0.1 km) for cross-source matching against the XLSX catalog.

3. Implemented Pipeline

3.1 Catalog merging

The merge script handles heterogeneous source schemas, builds the fuzzy deduplication key, performs cross-source deduplication, and outputs a single sorted CSV. It is fully documented and reproducible.

3.2 MISD and graph construction

The core pipeline implements the following stages:

- Catalog filtering by magnitude threshold, date range, and event cap
- MISD EM algorithm: initialises Gaussian temporal (σ_t) and spatial (σ_s) kernels and background rate μ; iterates E-step (compute w_{ij} weights) and M-step (re-estimate μ, σ_t, σ_s) until convergence
- Productivity term ρ(m) = 10^(b·(m − m_min)) with configurable b-value
- Directed graph construction using NetworkX: nodes = events (with time, lat, lon, depth, mag, background probability attributes); edges = $w_{ij} \geq$ threshold
- Edge labelling: direct vs. indirect (cascade) — an edge i→j is indirect if ∃ k such that i→k and k→j both exist
- Export to GraphML (Gephi/Neo4j compatible) and edge/node CSV
- Ego-network visualisation with node colour ∝ background probability, node size ∝ magnitude, edge colour by type

4. Preliminary Results: 2011 Tōhoku Sequence

As a proof-of-concept, I ran the pipeline on the M≥6.0 subset of the catalog centred on the 2011 Tōhoku earthquake (M9.1, 2011-03-11), with a magnitude cap of M≥4.5 and an event cap of 2,000 events. The EM algorithm converged within the configured iteration limit.

The resulting ego-network (2-hop neighbourhood around event 19, the M9.1 mainshock) is shown below. Key observations:

- The M9.1 node is the sole
- Secondary hubs (M7.9, M7.3) act as cascade sources, generating indirect (orange) edges to distal M6+ events — consistent with the known multi-segment rupture and early vigorous aftershock sequence
- Most leaf nodes are deep red (high background probability), indicating the MISD algorithm correctly identifies the majority of M6+ events in the broader catalog as background, not triggered by Tōhoku
- Direct (blue) edges fan radially from the mainshock hub; indirect (orange) edges form longer arcs through intermediate nodes, capturing the cascade structure

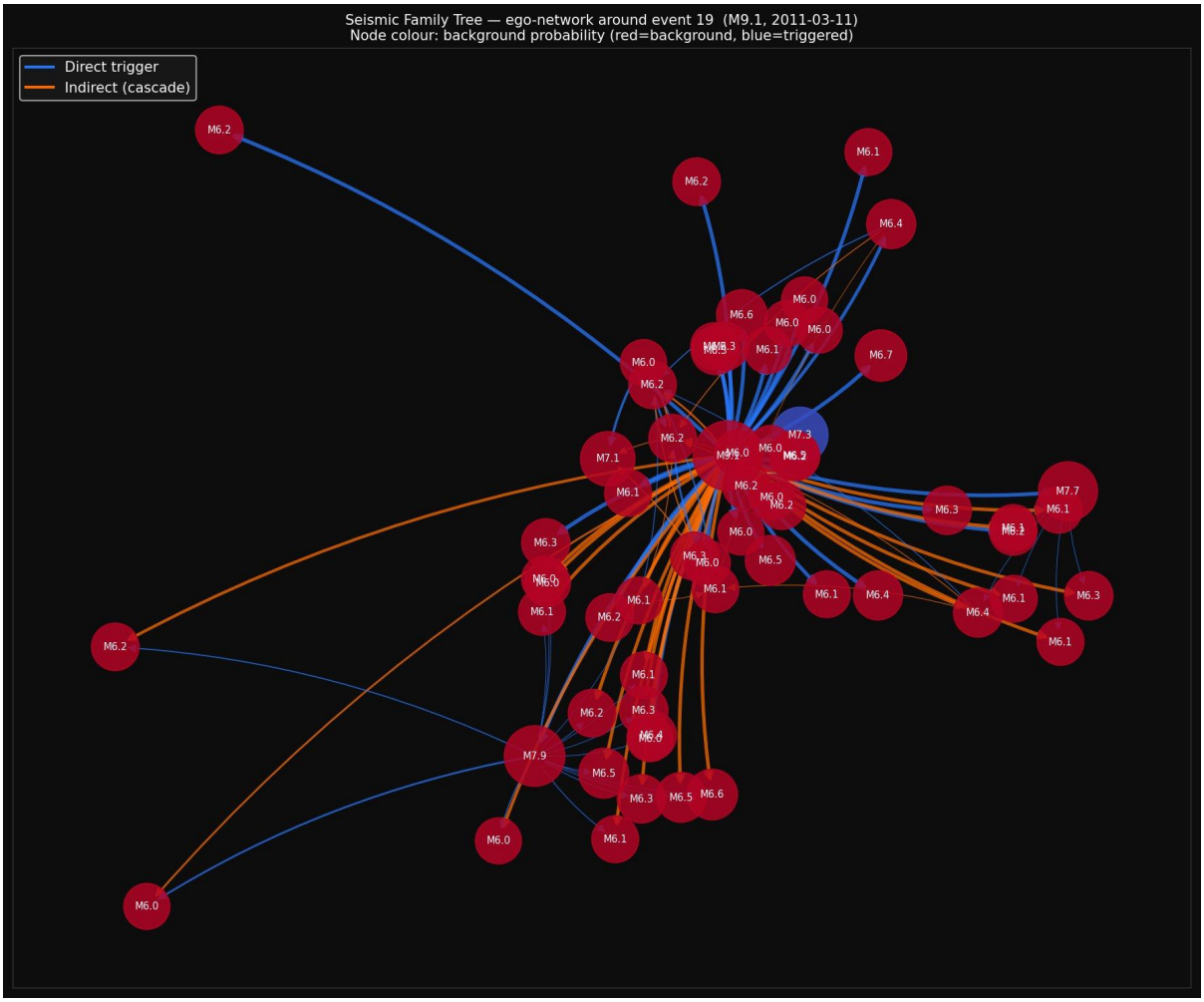


Figure 1. Ego-network (2-hop) around event 19 (M9.1, 2011-03-11 Tōhoku). Node colour: background probability (red = background, blue = triggered). Node size ∝ magnitude. Blue edges = direct triggers; orange = indirect cascade.

5. Proposed Extension as a MEXT Master's Student

The bachelor's thesis establishes the proof of concept: a working MISD-based graph construction pipeline applied to a merged Japan earthquake catalog (81,369 events, 1905–2025), with a preliminary ego-network result on the 2011 Tōhoku sequence. This builds directly on earlier independent research using the Baiesi–Paczuski nearest-neighbour method and the Zaliapin truncated space-time distance framework for seismic declustering — giving me hands-on experience with the statistical foundations before implementing the graph layer. The proposed master's research at Kyoto University would extend this work across four directions, each drawing on prior experience from the broader CS background documented in my study plan:

Direction	Description and prior background drawn upon
1. Application to Hi-net LFE catalogs	Apply the graph framework to high-resolution Hi-net low-frequency earthquake and tremor catalogs from the Nankai Trough — datasets with no public equivalent outside Japan. Prior experience with USGS multi-source catalog merging and deduplication provides a direct technical foundation for handling Hi-net's data formats.
2. Systematic topological	Compute branching factor, tree depth, Colless imbalance, and betweenness centrality distributions across multiple sequences;

characterisation	benchmark against a Galton-Watson branching null model; propagate uncertainty across stochastic reconstructions. The probabilistic modelling and graph algorithm skills developed through federated learning research and knowledge graph validation work are directly applicable here.
3. Cluster type discrimination	Test whether topological signatures distinguish mainshock-aftershock sequences from swarm-type clusters and slow from regular earthquake populations. Prior experience training classification models in federated and centralised settings provides a methodological basis for the supervised discrimination component.
4. Knowledge graph validation and corruption testing	Extend the knowledge graph validation framework developed in current project work — which extracts KGs, generates corrupted variants, and uses multi-agent architectures to iteratively validate and refine them — to the seismic domain. Corrupted graph variants (spurious edges, missing parents, inverted causality) would serve as a benchmark for testing the robustness of topological metrics and the reliability of the MISD reconstruction under catalog incompleteness.

A methodological comparison of ETAS-based reconstruction versus MISD — assessing the sensitivity of graph topology to modelling assumptions, and comparing both against the earlier Baiesi–Paczuski and Zaliapin declustering approaches used in prior work — would be conducted as a standalone contribution to the evaluation of existing frameworks.

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6. Summary

The core pipeline is implemented and produces scientifically coherent results on the Tōhoku sequence. The master's proposal is a natural and well-scoped extension of this work, focused on the Japanese data and seismological context that motivate studying in Japan specifically. I would be very glad to discuss any aspect of the methodology or results in more detail.

References

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